Let $R(t,s)$ be defined by

$$R(t,s) = c + a e^{b\frac{\vartheta(t)+\vartheta(s)}{2}} J_0\left(\beta|t-s|\right)$$

where a, b, and β are scaling parameters, c is a baseline adjustment, and J_0 is the Bessel function of the first kind of order 0, and $\vartheta(t)$ is the Riemann-Siegel theta function.

The integral covariance operator $\mathcal{K}$ associated with this kernel, acting on a function ϕ, is given by

$$(\mathcal{K}f)(t) = \int_0^\infty R(t,s)\, f(s)\, d\,s \tag{1}$$

so that

$$(\mathcal{K}\,\phi_n)(t) = \lambda_n \int_0^\infty \left(a e^{b\frac{\vartheta(t)+\vartheta(s)}{2}} J_0\left(\beta|t-s|\right) + c \right) \phi_n(s)\, d\,s \tag{2}$$

The eigenfunction equation for the integral covariance operator $\mathcal{K}$ is solved to yield characteristic eigenfunctions $\phi_n(t)$ such that

$$(\mathcal{K}\,\phi_n)(t) = \lambda_n \phi_n(t) \tag{3}$$

which expands to

$$\int_0^\infty \left(a e^{b\frac{\vartheta(t)+\vartheta(s)}{2}} J_0\left(\beta|t-s|\right) + c \right) \phi_n(s)\, d\,s = \lambda_n\, \phi_n(t) \tag{4}$$